

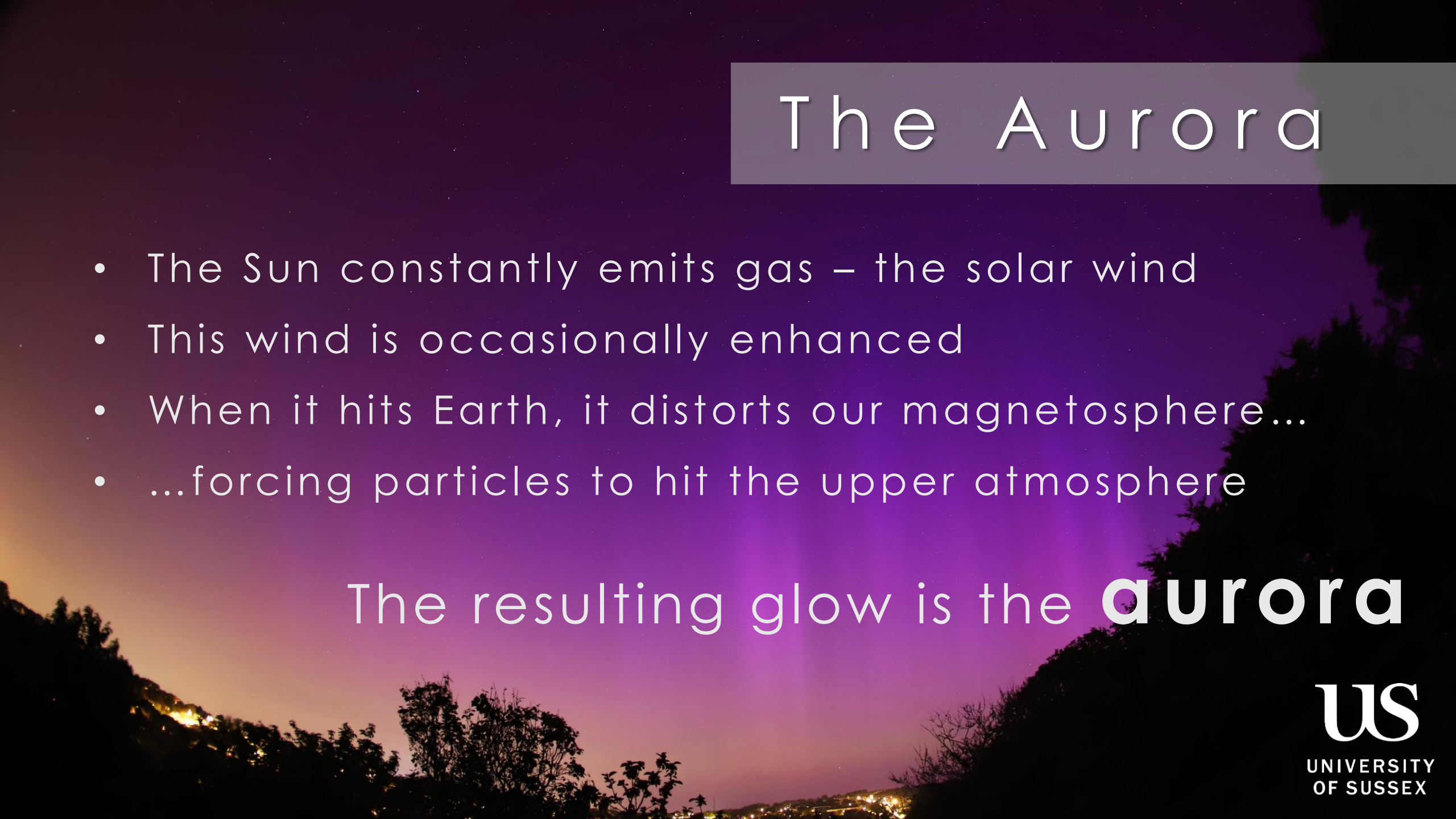
The Aurora

Dr Das Baskill
University of Sussex

The Aurora

- The Sun constantly emits gas – the solar wind
- This wind is occasionally enhanced
- When it hits Earth, it distorts our magnetosphere…
- …forcing particles to hit the upper atmosphere

The resulting glow is the **aurora**



UNIVERSITY OF SUSSEX





The Sun



White light photograph of the Sun, Saturday 17th October, 2026

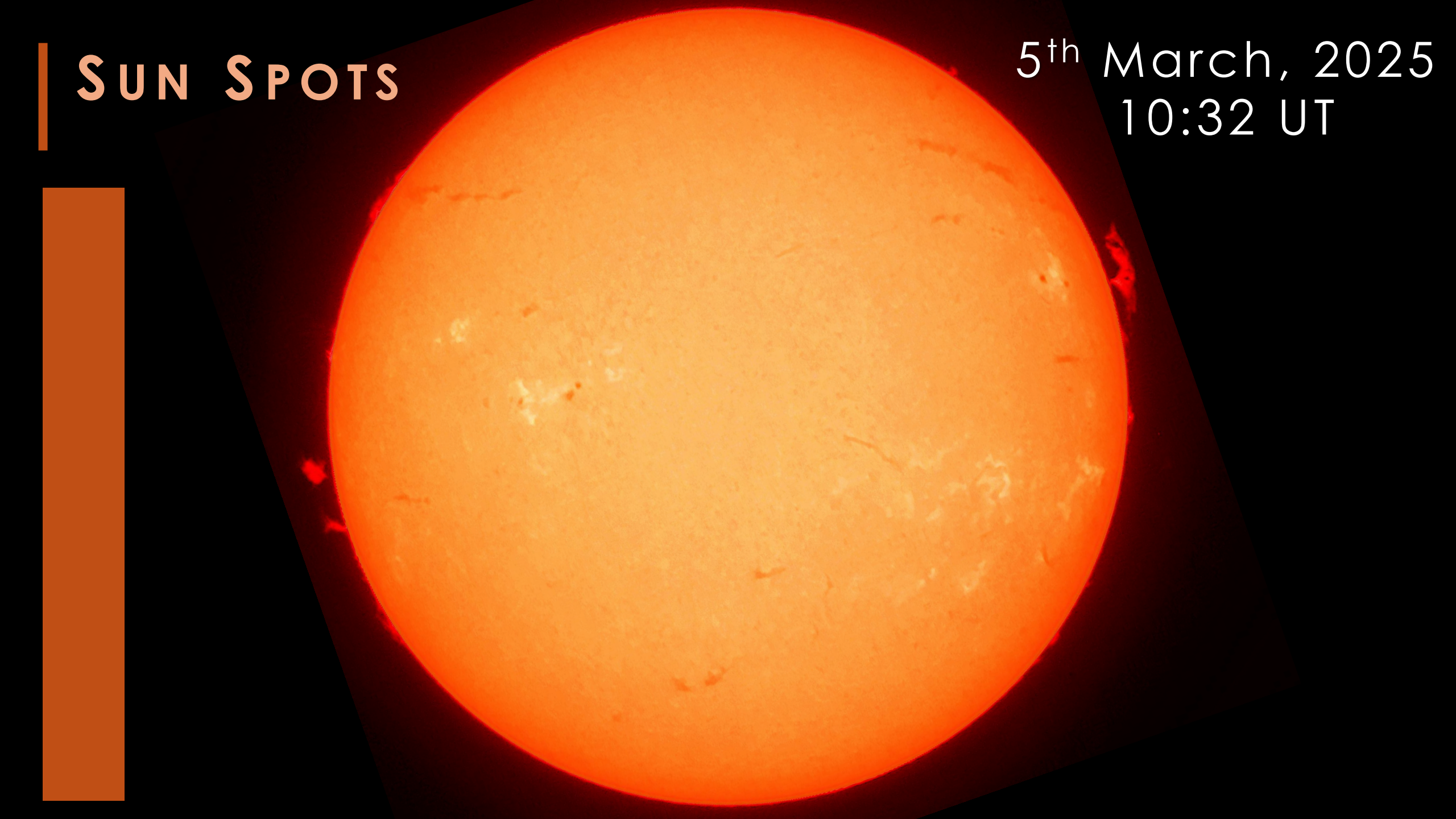
The Sun



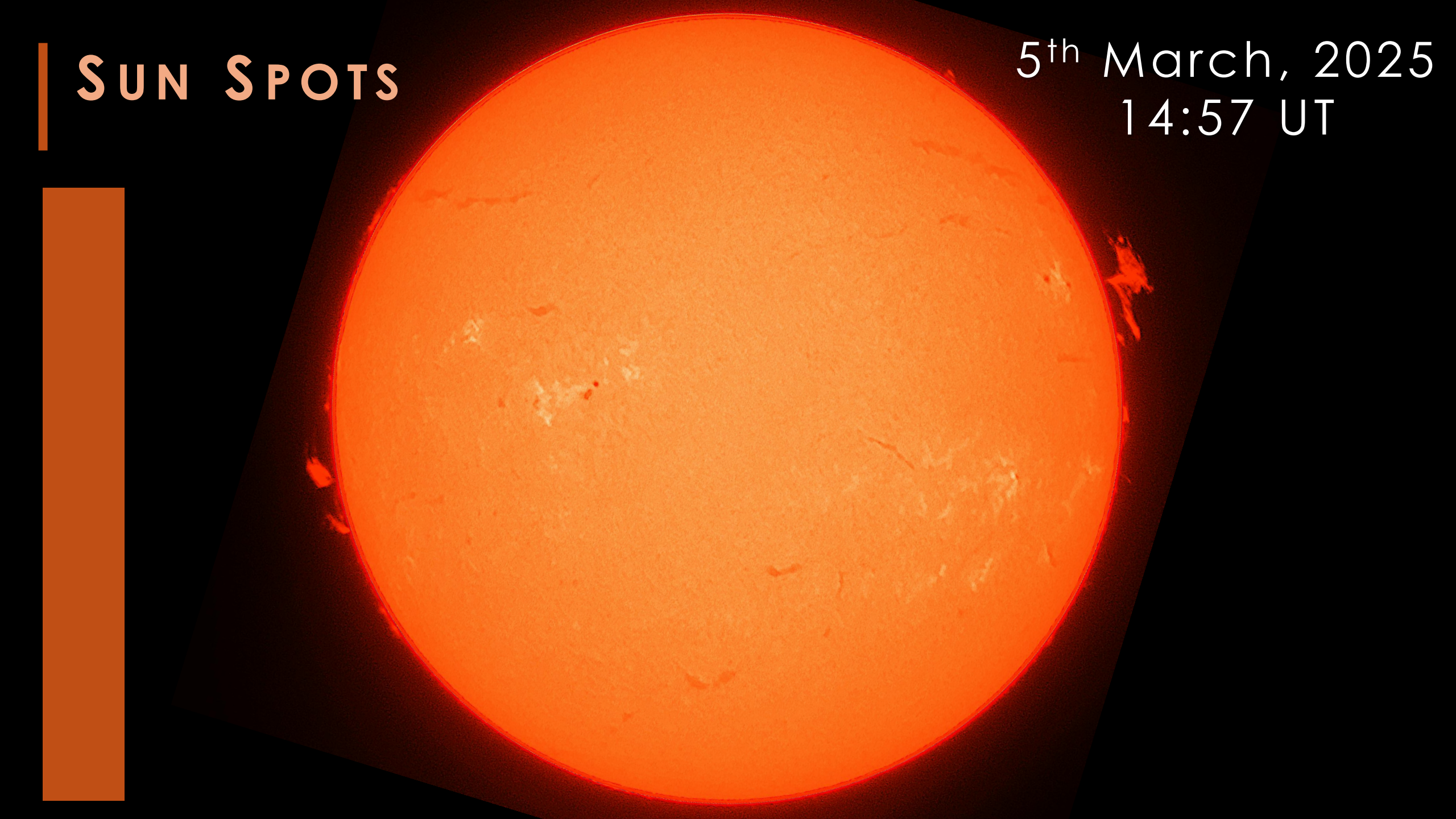
H-alpha view
of the Sun
4th August, 2024

Sun Spots

5th March, 2025
10:32 UT



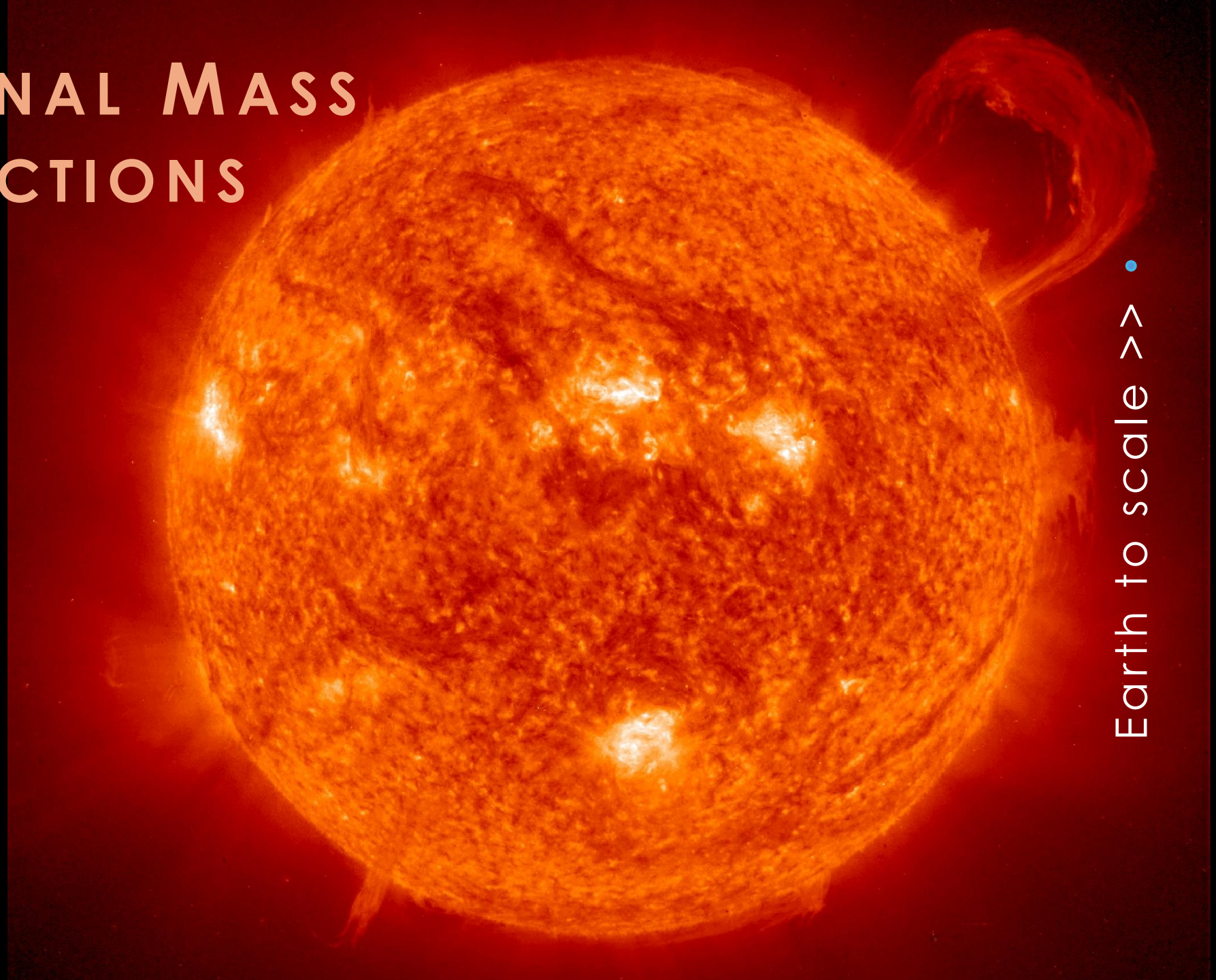
Sun Spots



5th March, 2025
14:57 UT



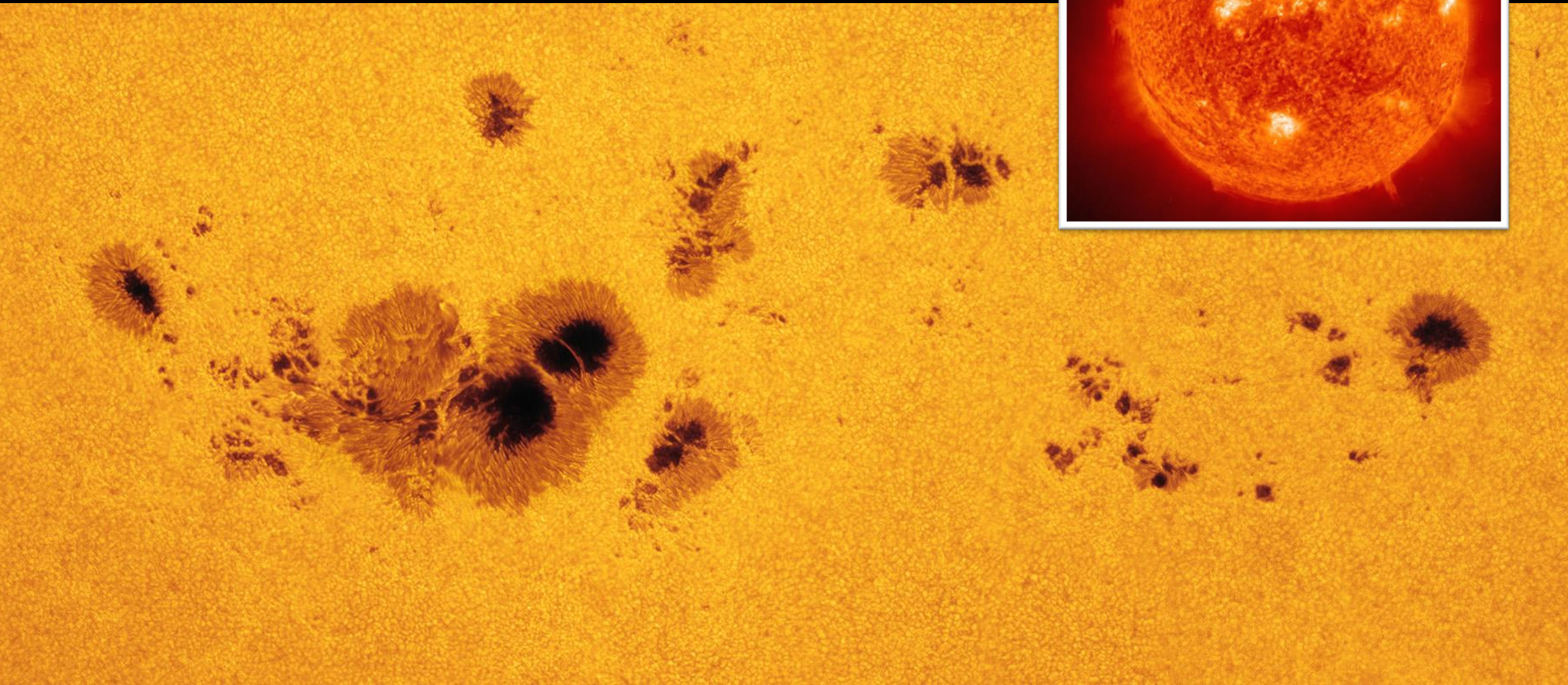
Coronal Mass Ejections

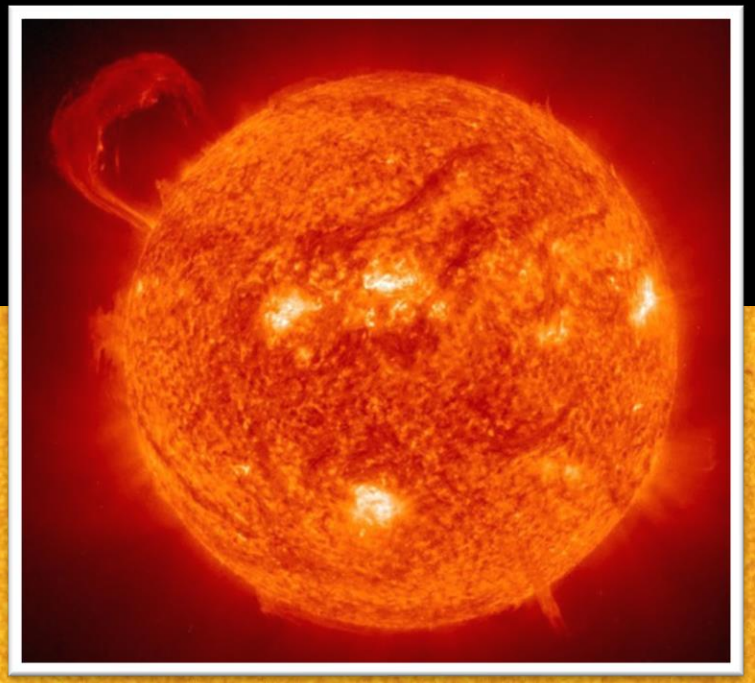


Earth to scale >>

SOHO Solar Observatory (Nasa/Esa)

Sun Spots

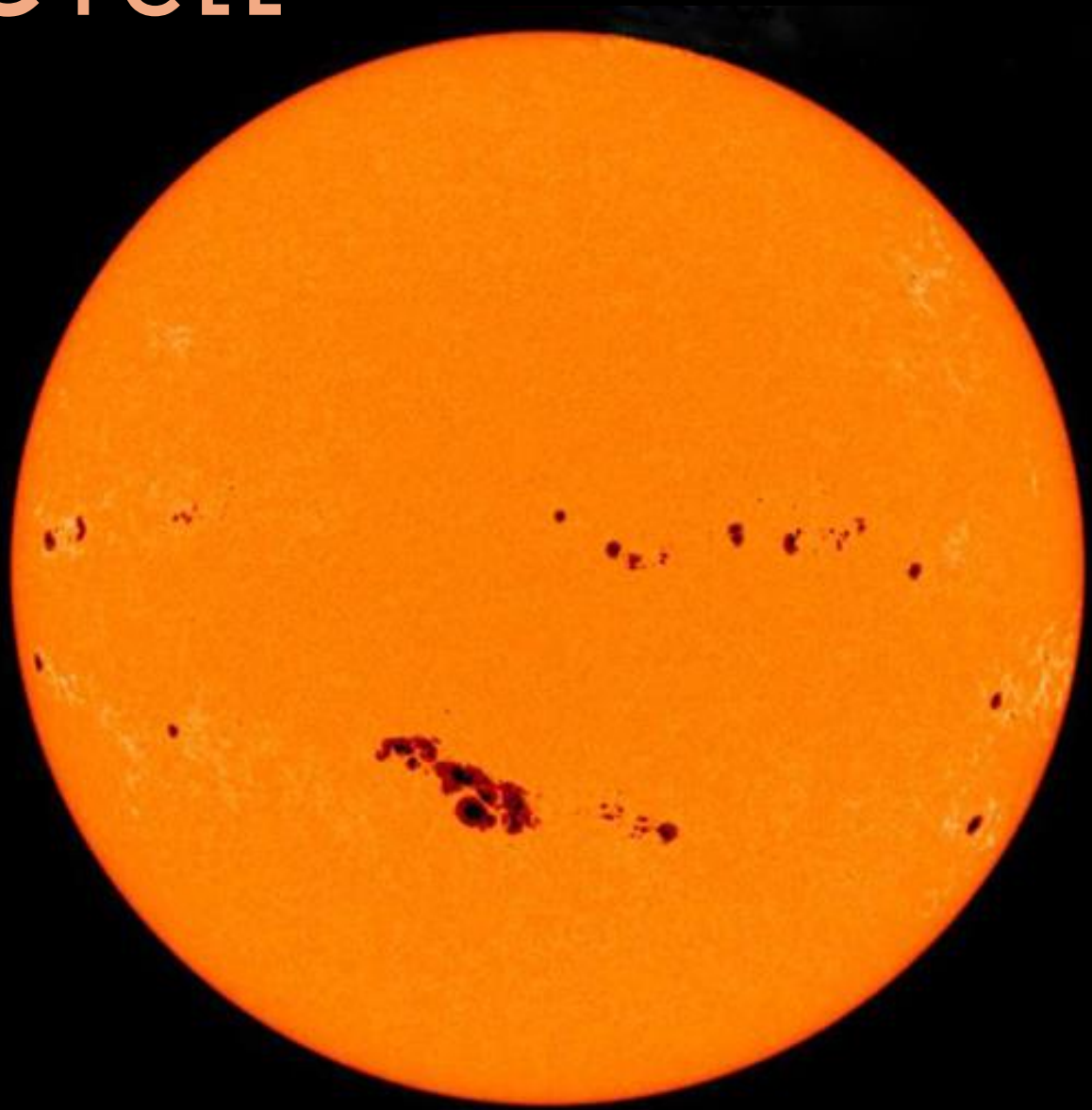




The 11-year Solar Cycle

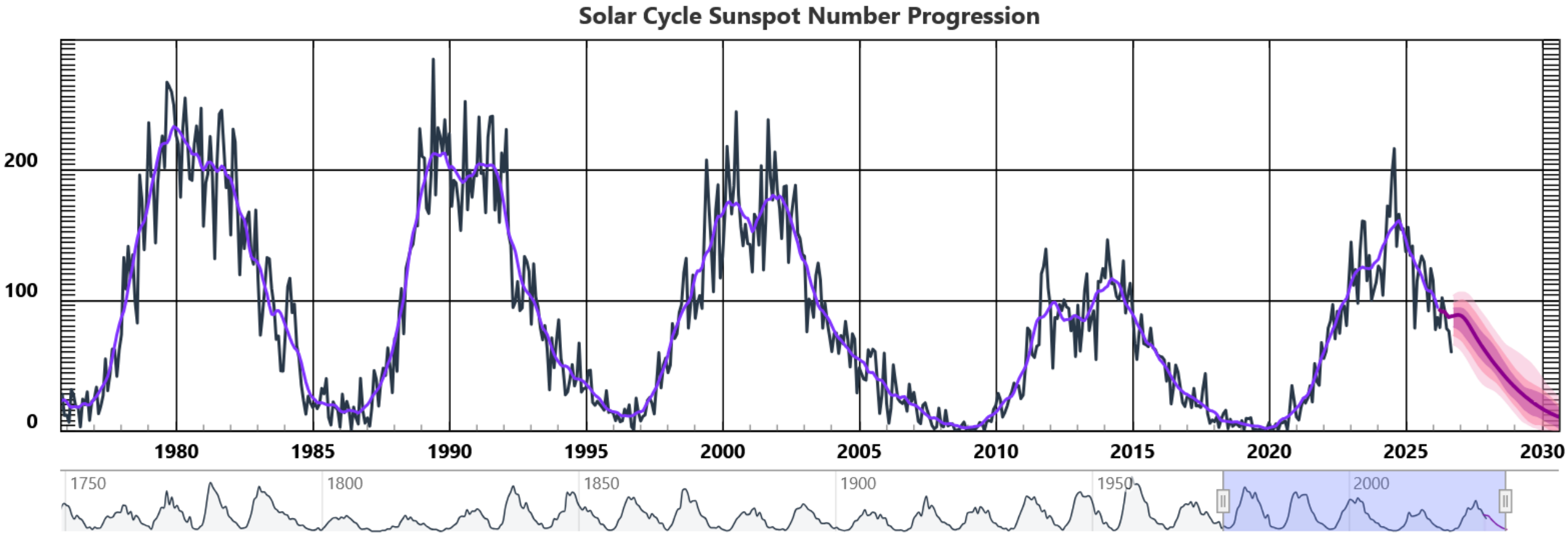


Solar Minimum



Solar Maximum

Solar Cycle


Solar Cycle Sunspot Number Progression
200
100
0
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2030
1750
1800
1850
1900
1950
2000

Sun Spots

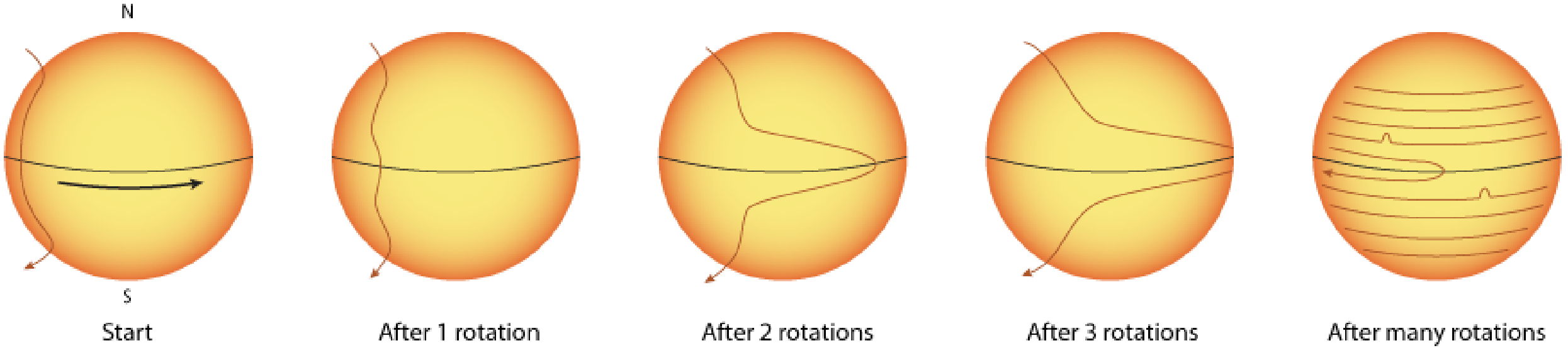
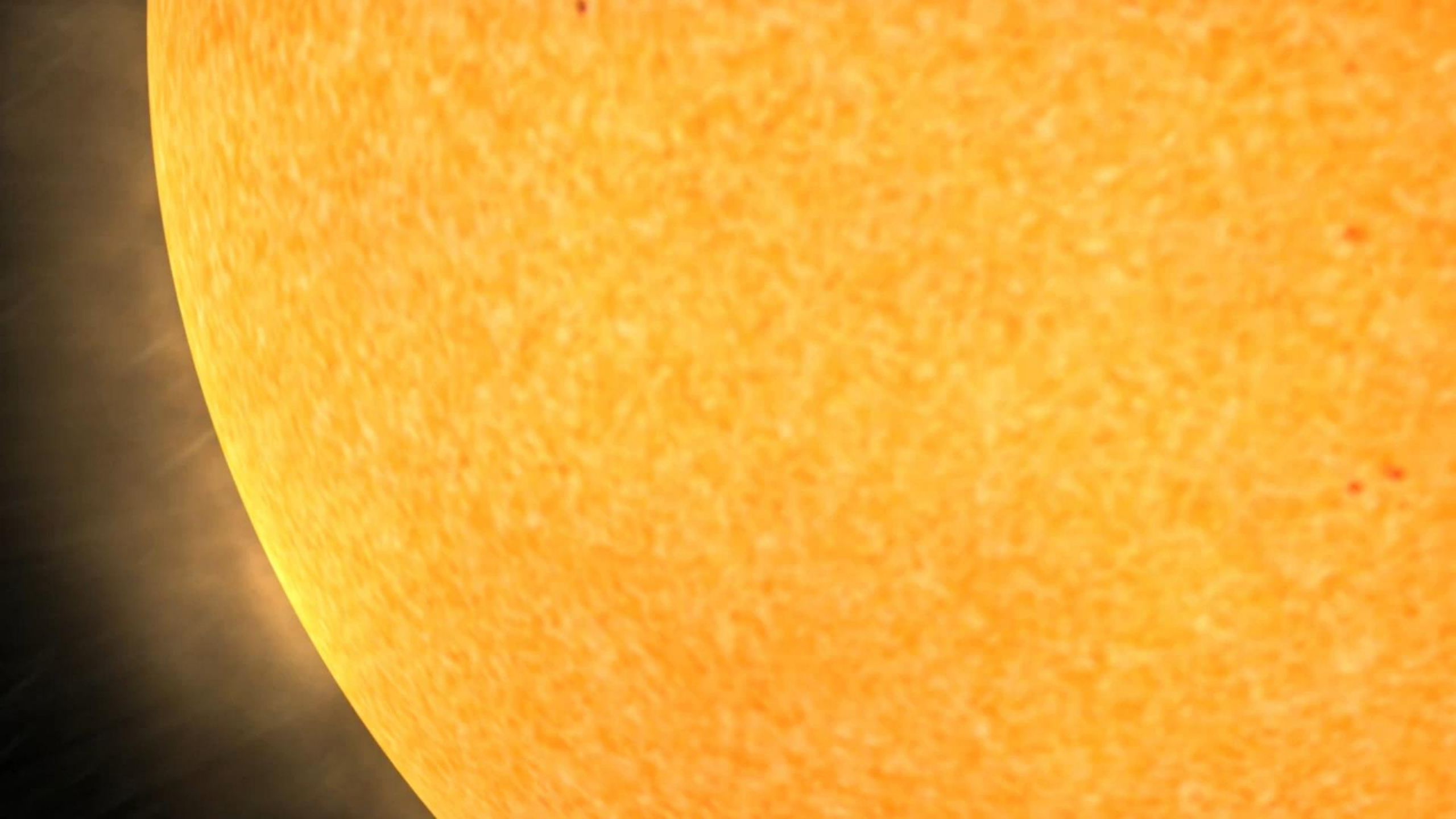
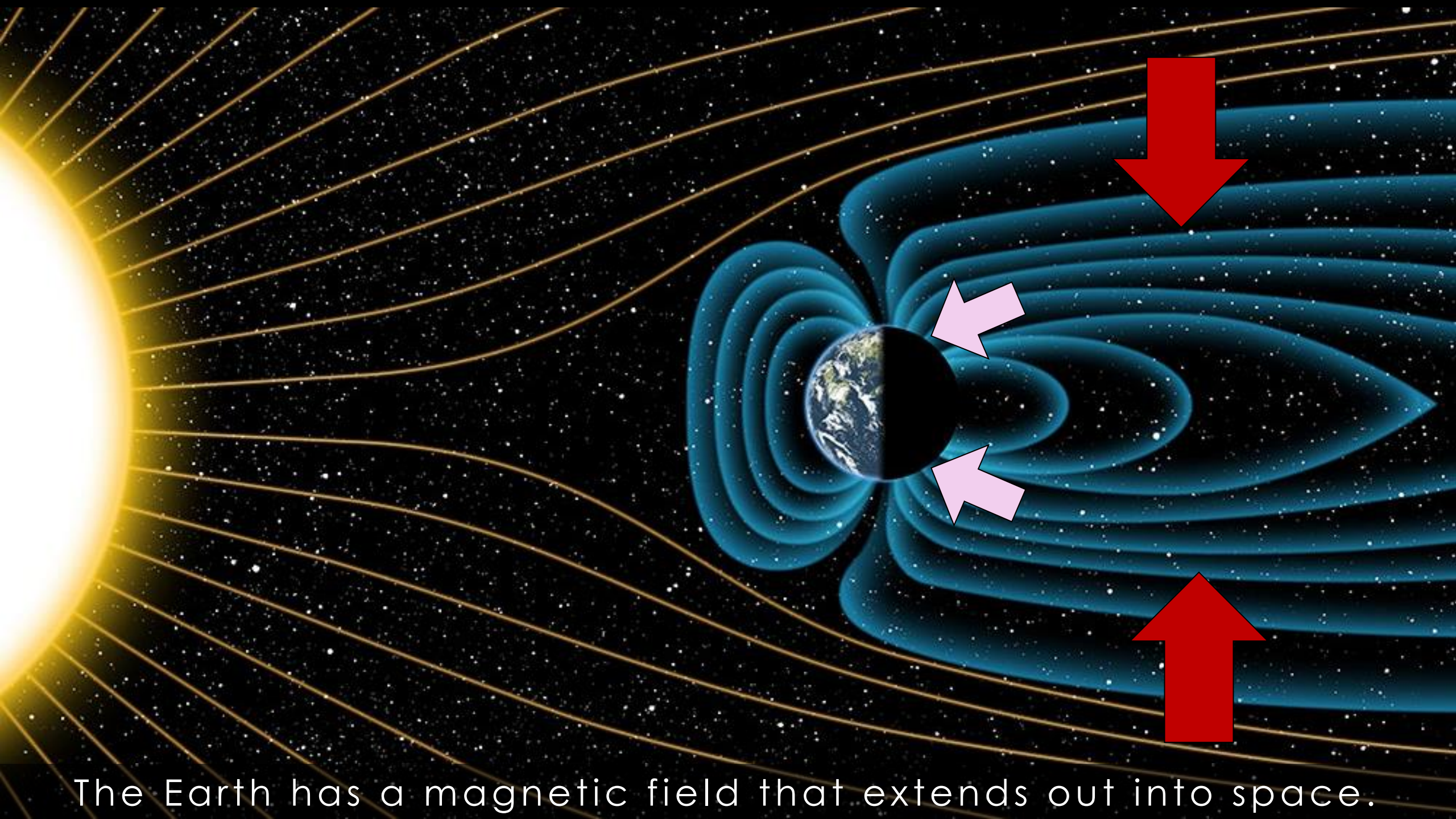
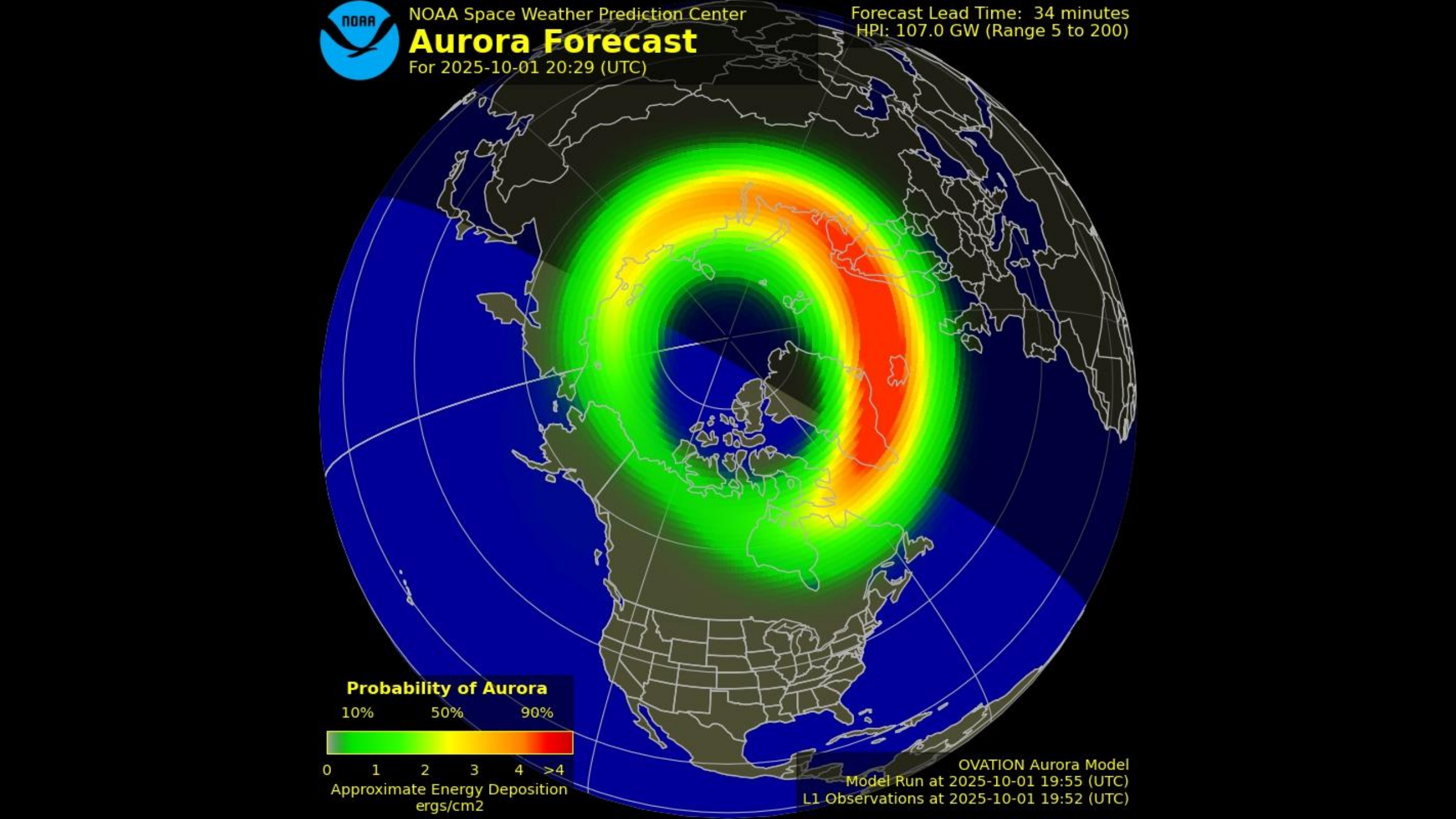


Illustration by José Francisco Salgado, PhD (Adler Planetarium)





The Earth has a magnetic field that extends out into space.


NOAA Space Weather Prediction Center
Aurora Forecast
For 2025-10-01 20:29 (UTC)
Forecast Lead Time: 34 minutes
HPI: 107.0 GW (Range 5 to 200)
Probability of Aurora
10%
50%
90%
0
1
2
3
4
>4
Approximate Energy Deposition
ergs/cm2
OVATION Aurora Model
Model Run at 2025-10-01 19:55 (UTC)
L1 Observations at 2025-10-01 19:52 (UTC)


August 2014 over Greenland

The Colours of the aurora



Red

High-altitude oxygen >200km, visible during strong solar activity.

Green

Common due to the abundance of Oxygen between 100-200 km.

Pink, purples & blues

Nitrogen molecules emit red and blue colours when energetic particles reach lower altitude.



Aurora Types

Only really two types…

- Discrete (Arcs, curtains, etc)
- Diffuse (patchy, pulsating, etc)

But these can look very different from different angles.

Aurora Types



Discrete:
Arcs, bands, curtains, pillars, rays, etc

Aurora Types

Diffuse:

Corona, Patches, etc



Aurora Types

Diffuse:

Corona, Pulsating, Patches, etc

Aurora Types



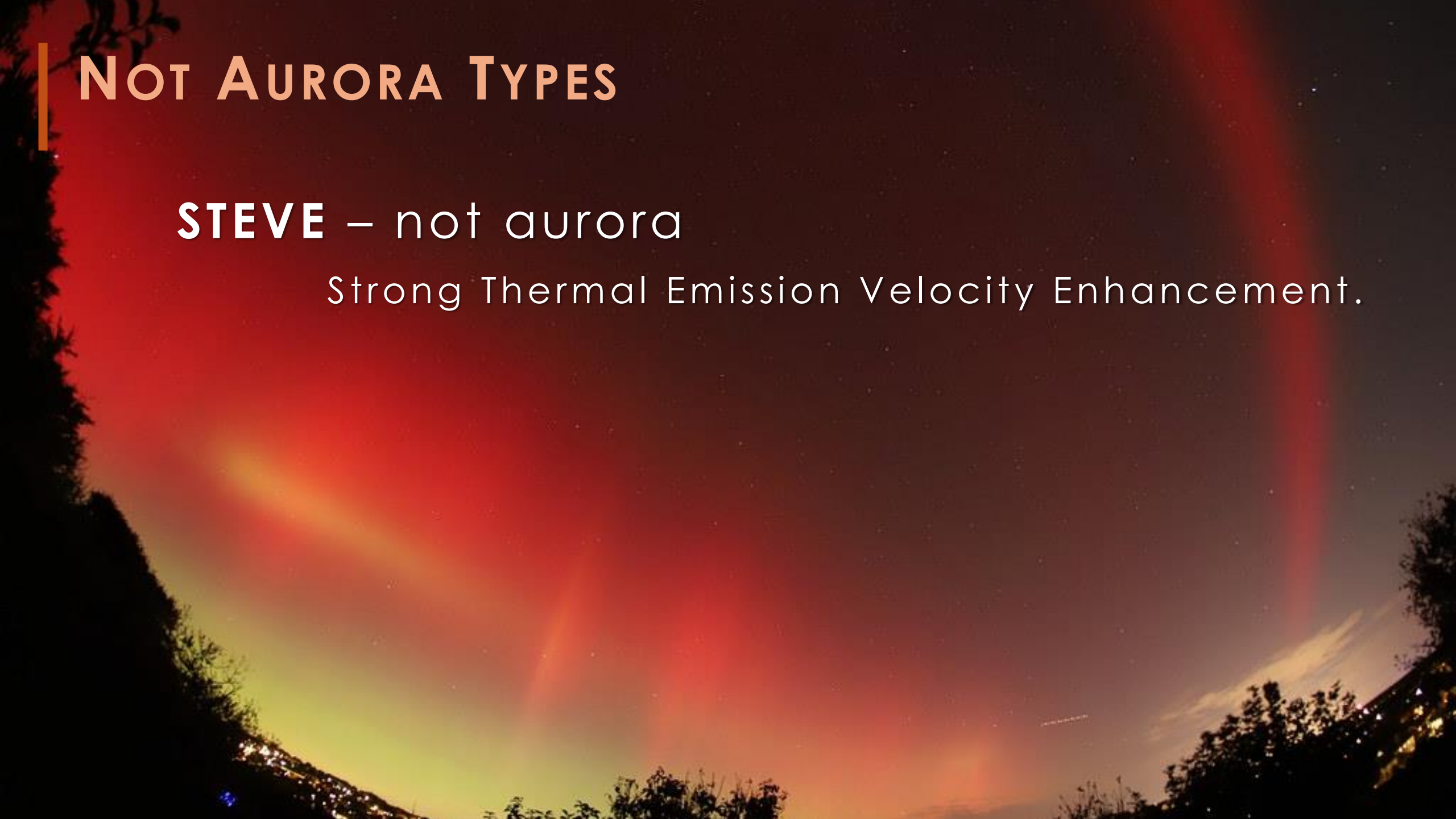
Diffuse:

Corona, Pulsating, Patches, etc

Not Aurora Types

STEVE – not aurora

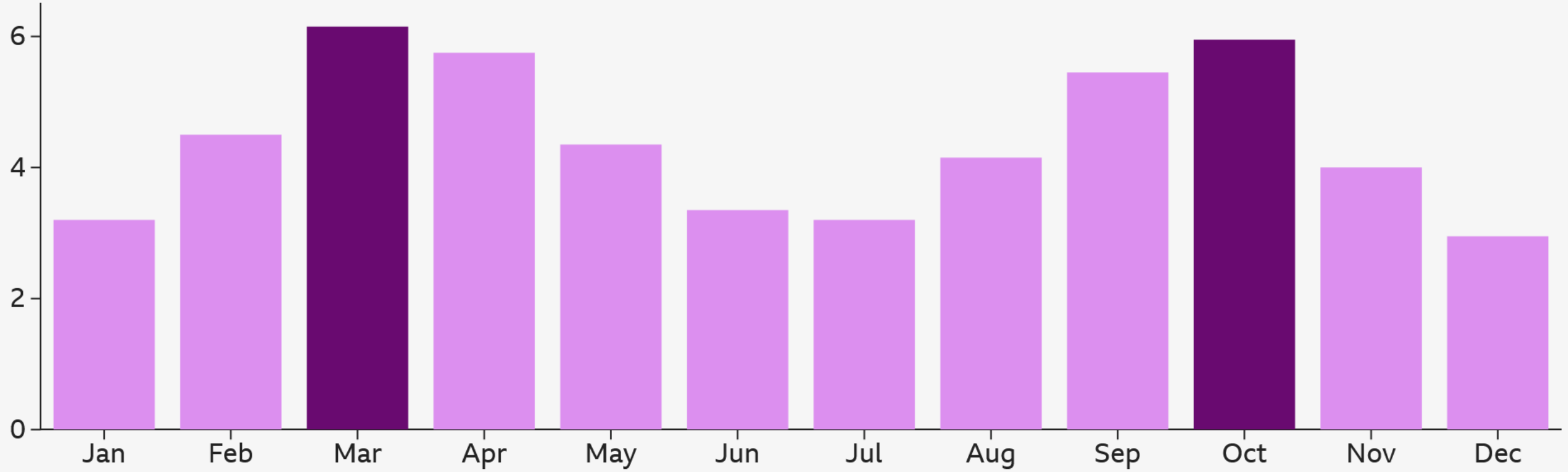
Strong Thermal Emission Velocity Enhancement.



Stronger aurora more likely in spring and autumn

Effects are caused by 'geometric disturbances' - which increase around **equinox months**

Average number of geometrically disturbed days each month 1932-2007



Source: Nasa physicist David Hathaway

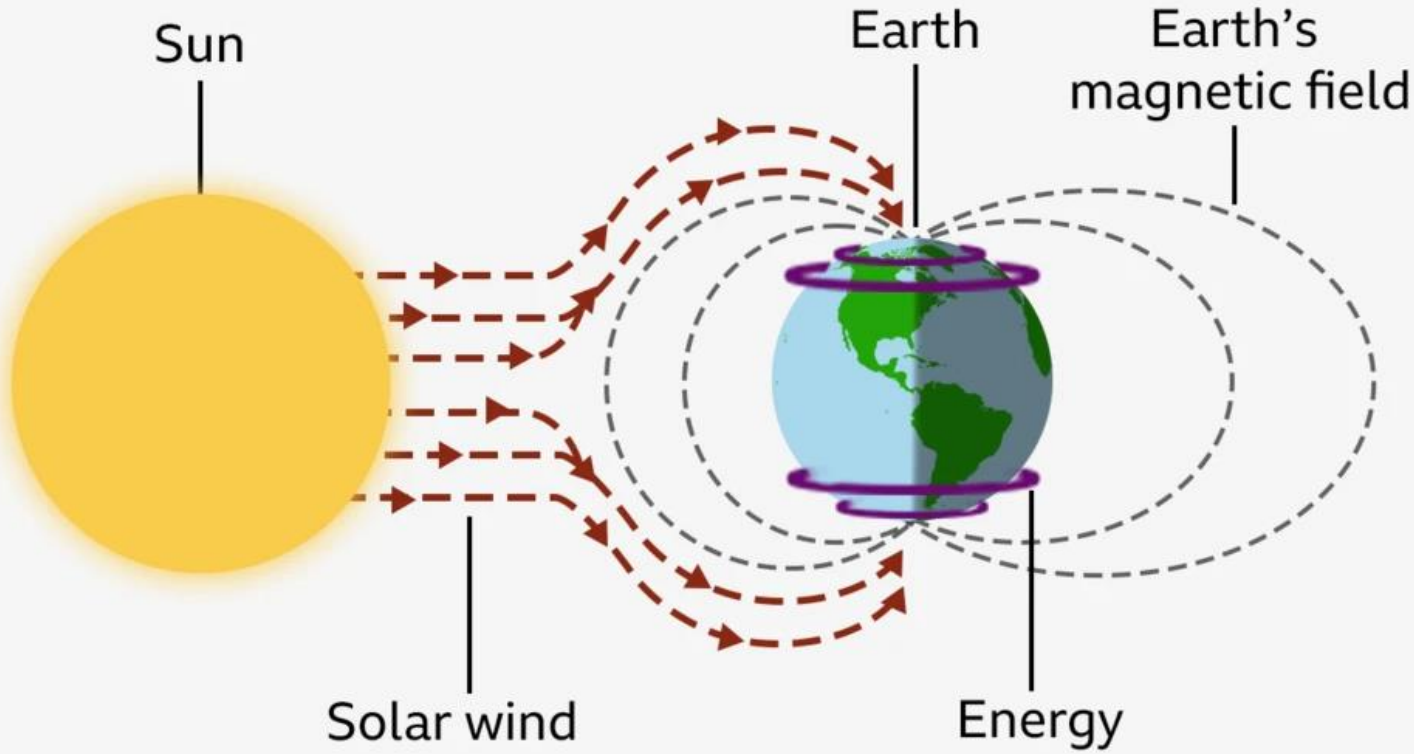
BBC

Russell-McPherron effect, due to the Earth's tilt

Equinoxes

How the Russell-McPherron effect can make the aurora stronger

When there is no tilt - at **equinox -** the magnetic field of the solar wind is more attracted to the Earth's magnetic field - resulting in more energy entering the atmosphere and **stronger aurora**



Source: BBC research/Noaa

BBC

WEBSITES

https://www.gi.alaska.edu/monitors/aurora-forecast

AURORA FORECAST

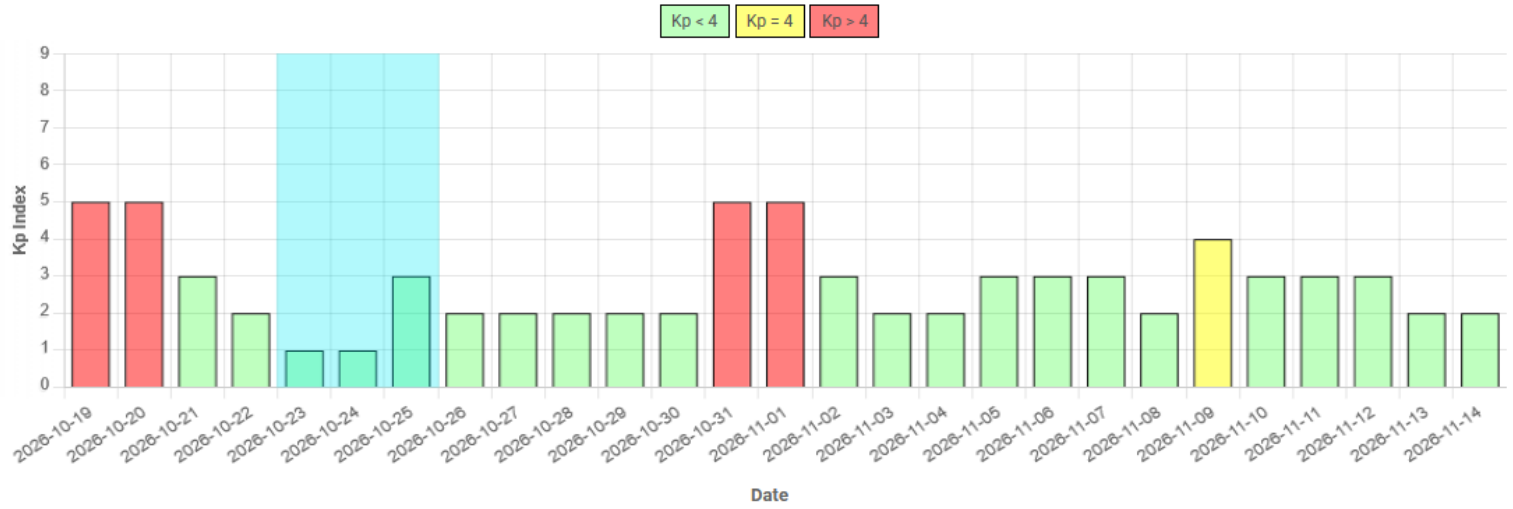
< Previous Day | Current | Next Day >

You are viewing 3-Day Forecast data in Alaska time.

Tuesday, March 25, 2025

3-Day Forecast — More Reliable | 27-Day Forecast — Less Reliable | Historical Data — Actual Activity

Time Zone: Local (British)



Data source: We pull data from the NOAA 27-Day Space Weather Outlook every Monday at 16:00 PM (British). The Space Weather Outlook is the forecast of geomagnetic activity in 27 days. NOAA updates the 27-Day Space Weather Outlook every Monday at 15:00 UTC.

The data: Highlighted in blue are 3-Day Forecast data overlaying the 27-Day Forecast. 3-Day Forecast data are updated daily and provides a more accurate forecast.

Last updated: 2026-10-19 16:00 PM (British).

FAQ

Read our FAQs to learn about these common questions and more:

Viewing the Aurora

- What time of year is best for seeing aurora?
- Can I see aurora during the summer months?
- Where is the best place to see aurora?
- Where should I view the aurora when I'm in Fairbanks, AK?

Science of the Aurora

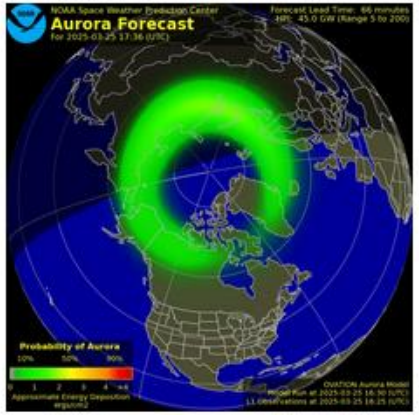
- What are Kp numbers?
- What does the 2024-25 solar maximum mean for aurora?
- Can you predict where there will be aurora?

Forecasting the Aurora

- How accurate is the 3-day forecast?

Current Activity

Watch the last few seconds of the animation to see what is predicted over your location in the next 30 minutes.



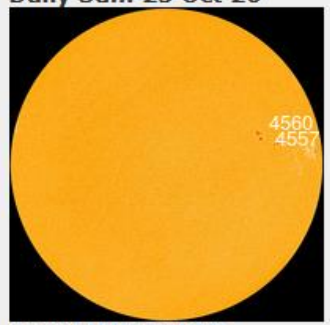
Source: Space Weather Prediction Center

Current Conditions

Solar wind
speed: **418.1** km/sec
density: **4.2** protons/cm^3
more data: ACE, SOLAR-1
Updated: Today at 1308 UT

X-ray Solar Flares
6-hr max: **C1** 1001 UT Oct23
24-hr: **C3** 0542 UT Oct23
explanation | more data
Updated: Today at: 1311 UT

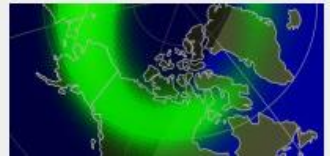
Daily Sun: 23 Oct 26



Expand: labels | no labels
Sunspot 4560 has a 'beta-gamma-delta' magnetic field that harbors energy for X-class solar flares. Credit: NASA/SDO

Sunspot number: 49
What is the sunspot number?
Updated 23 Oct 2026

Current Auroral Oval:



What's up in space

Friday, Oct. 23, 2026

AI writing is everywhere -- except here. This website is written by Dr. Tony Phillips, a carbon-based lifeform with 30 yrs of forecasting experience. Our guarantee: If you find a mistake, it was made by a real human being.



A POTENTIALLY DANGEROUS SUNSPOT: Sunspot 4650 has developed a 'beta-gamma-delta' magnetic field that harbors energy for X-class solar flares. Any eruptions today will likely be geoeffective because the sunspot is within 45 degrees of the sun-Earth line. **Solar flare alerts:** SMS Text

INFERIOR CONJUNCTION OF VENUS: Venus is passing almost directly between Earth and the sun. Astronomers call this "the inferior conjunction of Venus," and it's happening right now, Oct. 23-24. This is what the planet looks like:



"The delicate crescent of Venus was a lovely sight," says photographer

archives
October
23
2026
view

SATELLITE DRAG
today: **55.5** m/day ↓



2026 max: **127.6** (Jan 20)
2026 min: **19.7** (Aug 10)

Explanation: This is how fast a typical Starlink at 480 km is sinking due to aerodynamic drag: more information.
Updated: 23 Oct 2026

Starlink Statistics
Total Launched: 13,015
In orbit: 11,154
Updated: 22 Oct 2026

Starlink Re-entries (2026)
January: 24
February: 54
March: 48
April: 51
May: 60
June: 35
July: 52
August: 32
September: 47
October: 36
Updated: 22 Oct 2026

Starlink Re-entries (yearly)
2020 total: 46
2021 total: 78
2022 total: 100
2023 total: 88
2024 total: 308

See the aurora borealis from the UK

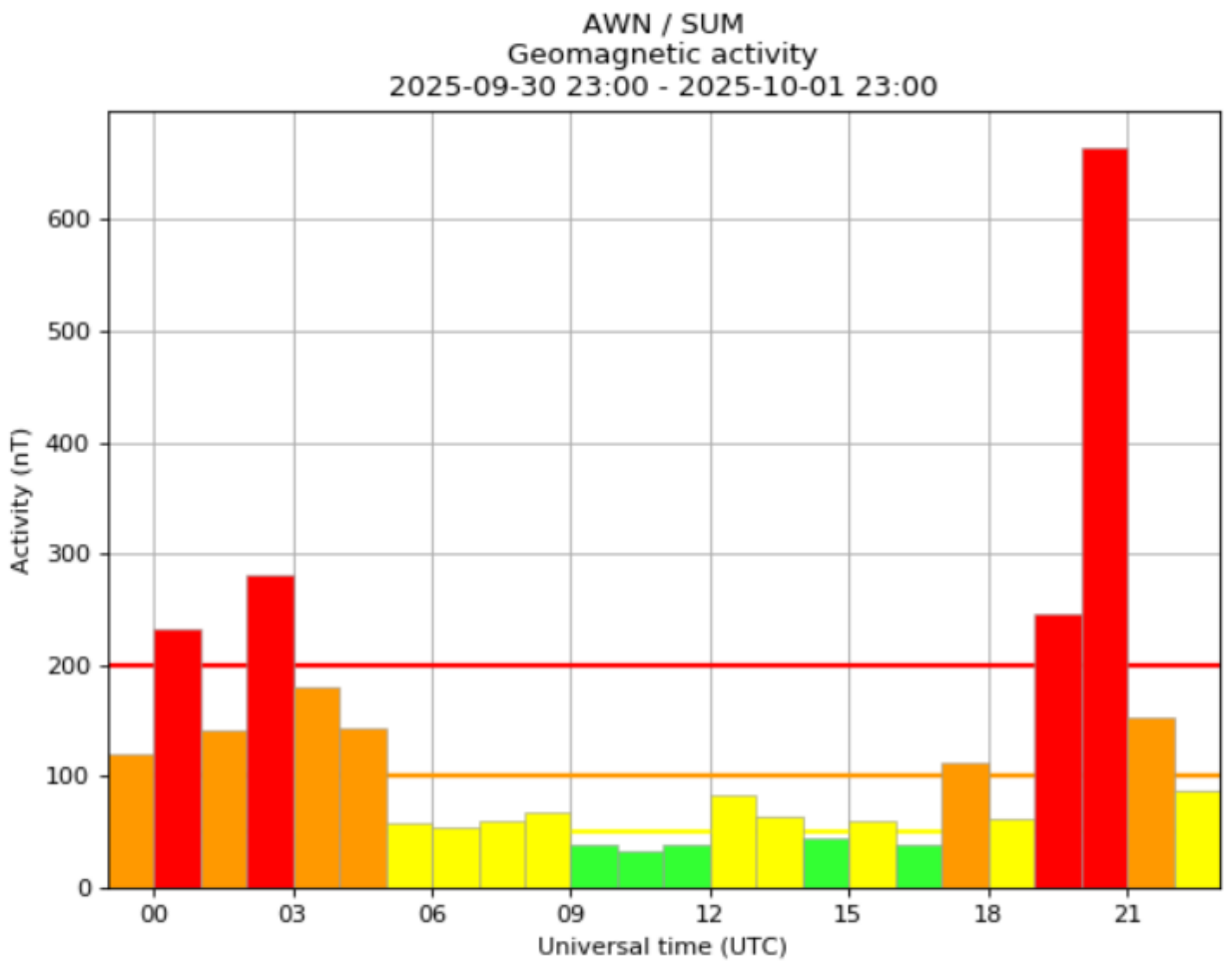
AuroraWatch UK

- Home
- Alerts
- Shetland alerts
- Plots
- Map
- Build
- FAQs
- Contact Us
- Blog
- Twitter
- Telegram
- Facebook
- Flickr
- Instagram
- YouTube

Lancaster University

AuroraWatch UK status:

Minor geomagnetic activity



Activity plot | Magnetogram | Stackplot

For Sussex...

500nT
– visible to cameras

750nT
– potentially visible by eye

1000nT
– easy to see by eye

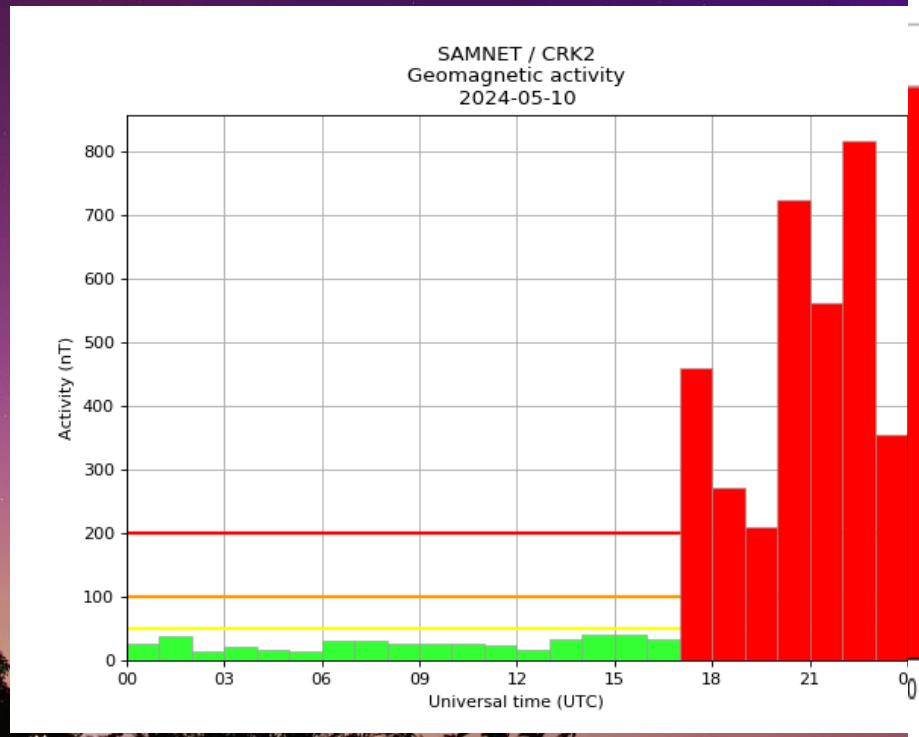
https://aurorawatch.lancs.ac.uk/

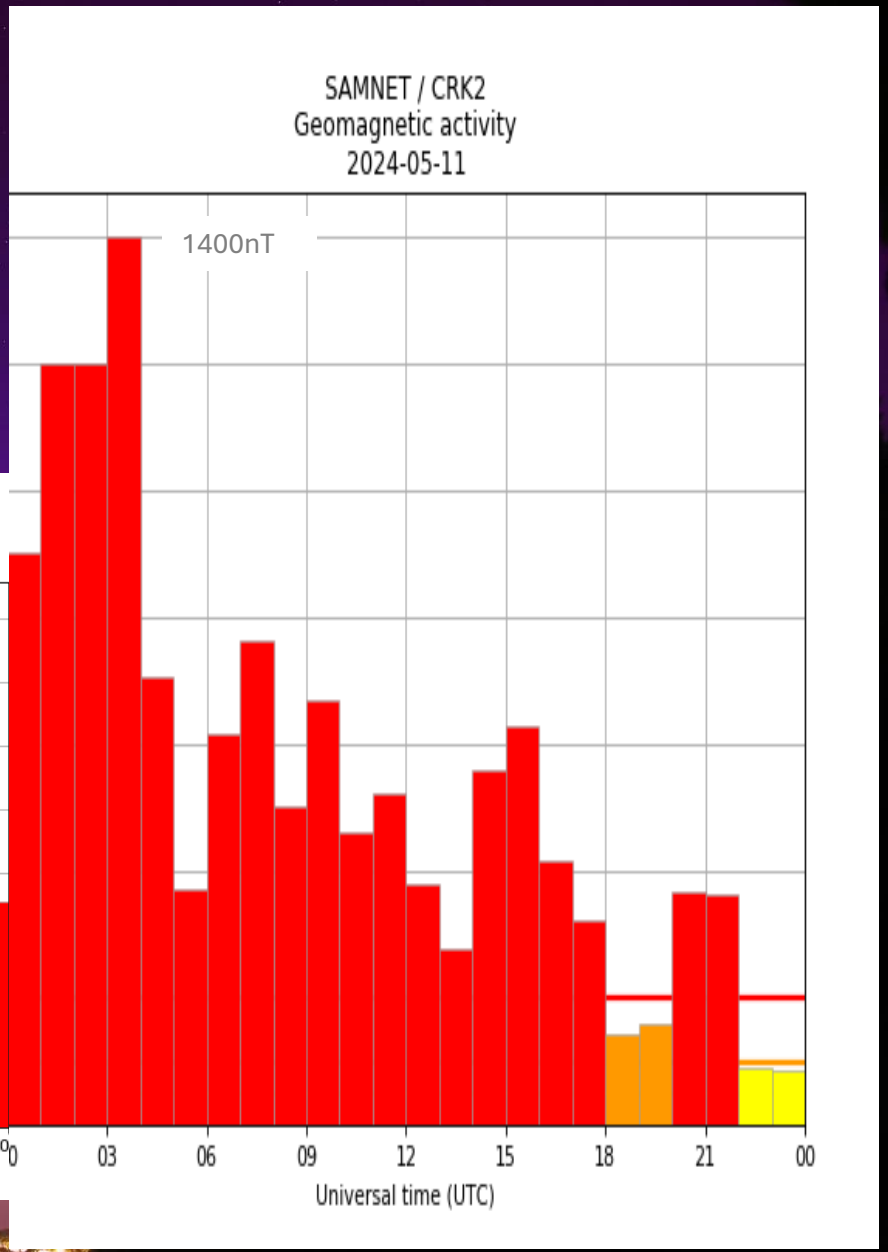
© AuroraWatch UK 2015.

Recent high aurora activity



Brighton – May 2024





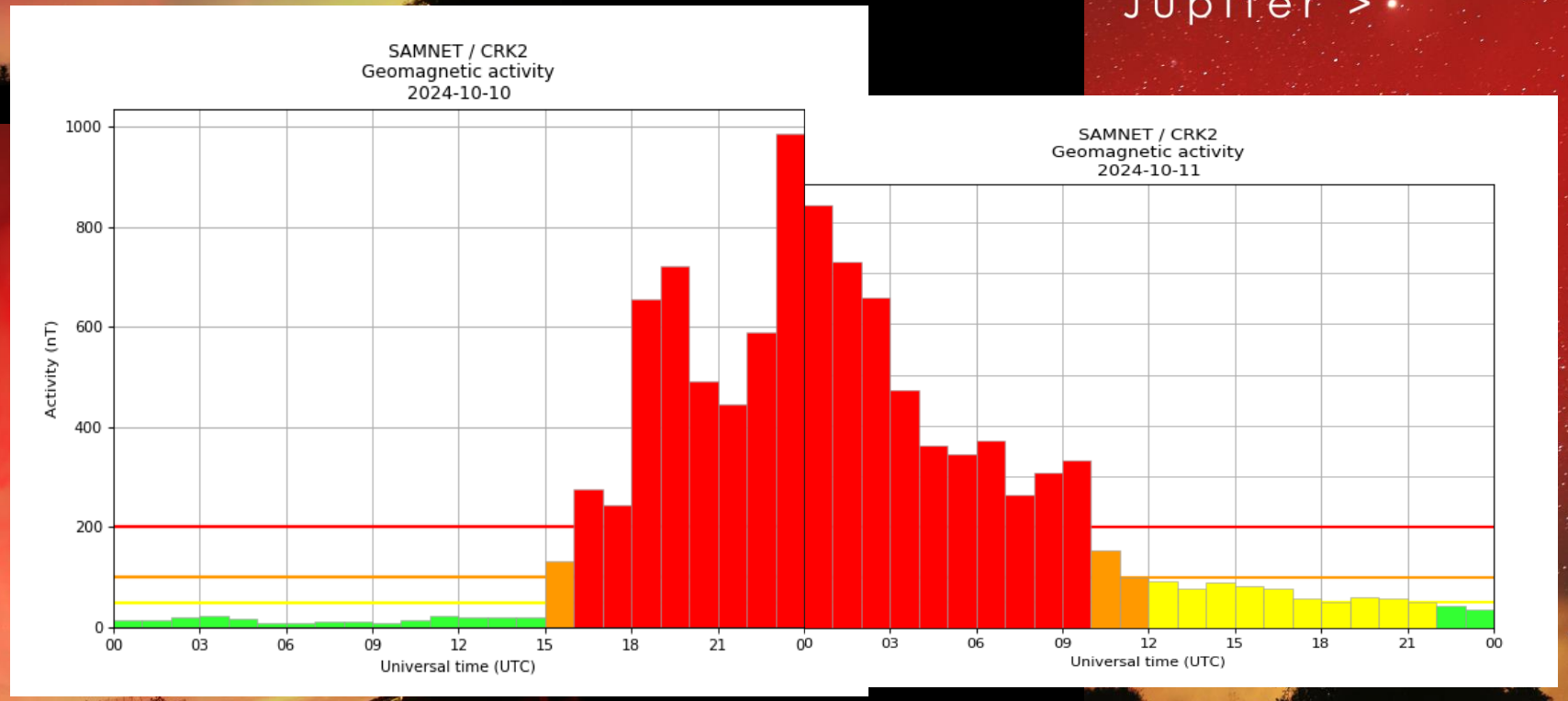
Brighton – October 2024

Pleiades >

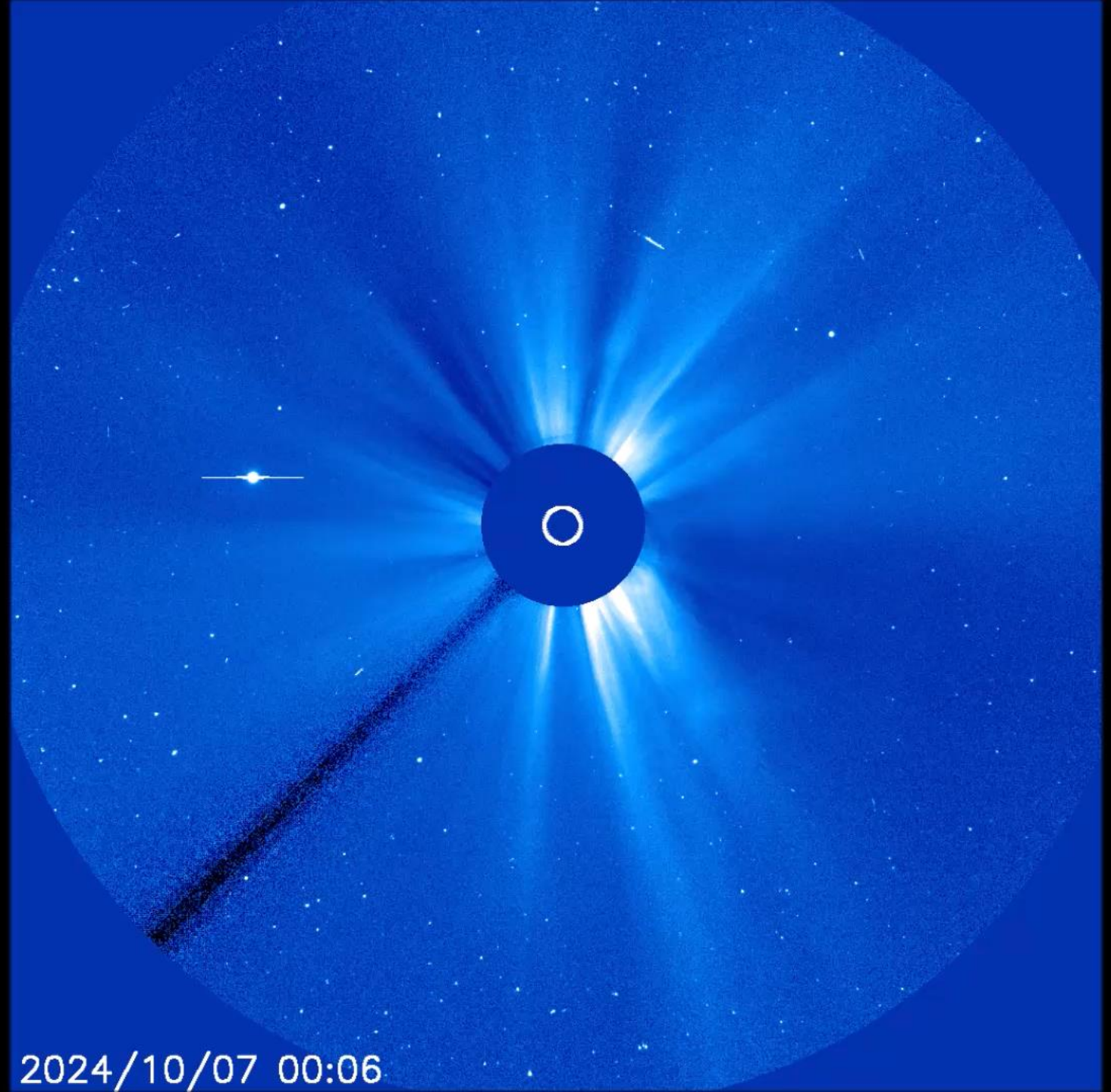
Hyades

Jupiter >

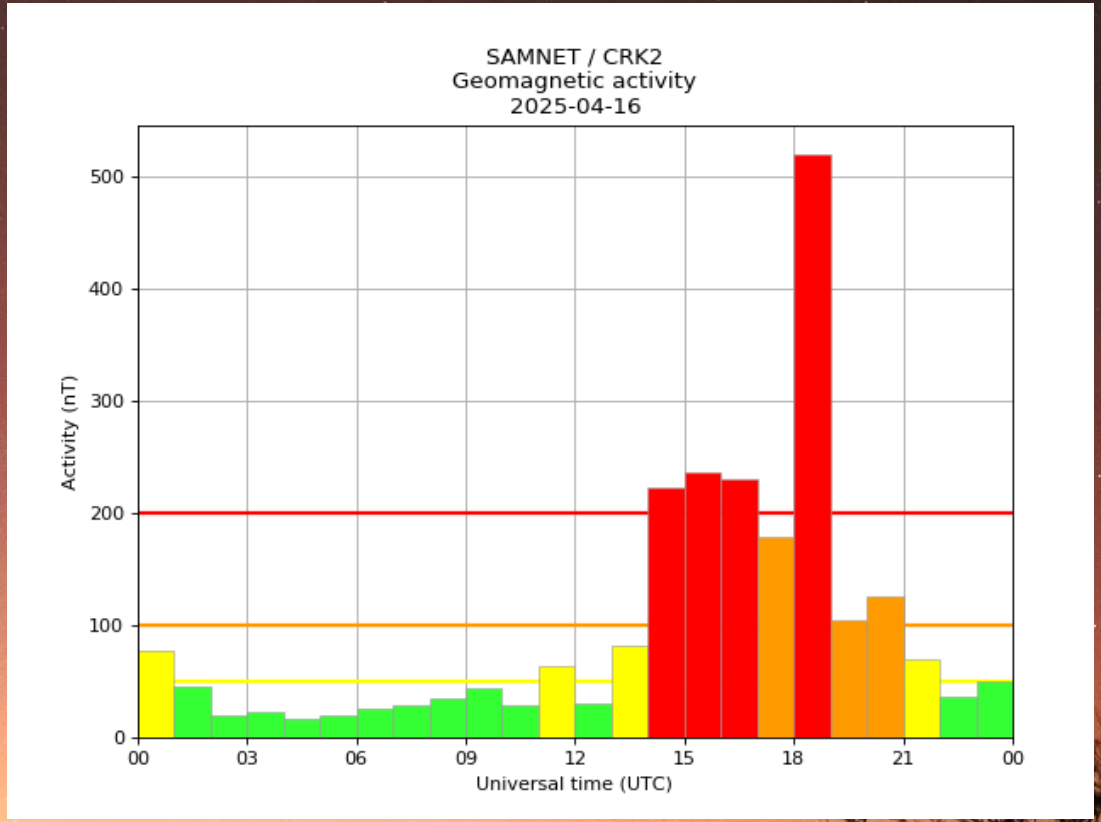
Orion



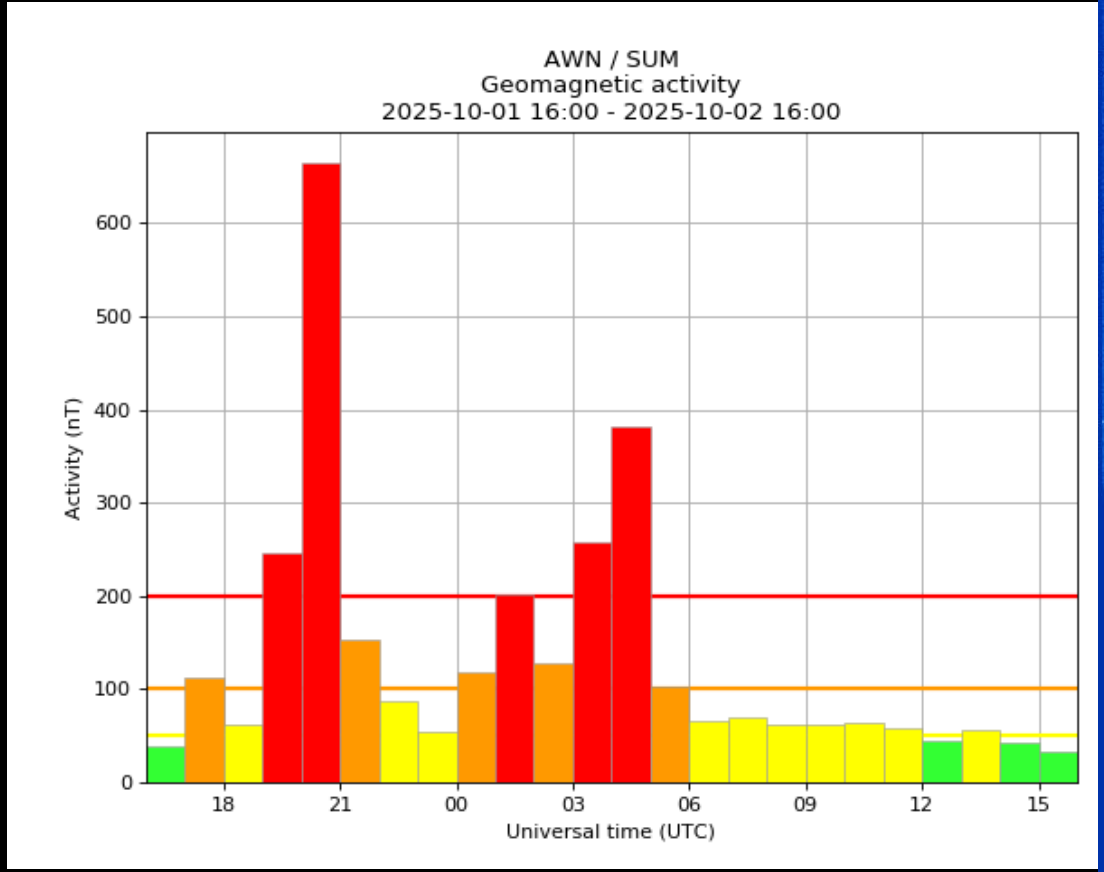
The Solar activity that sparked aurora over southern England in October 2024

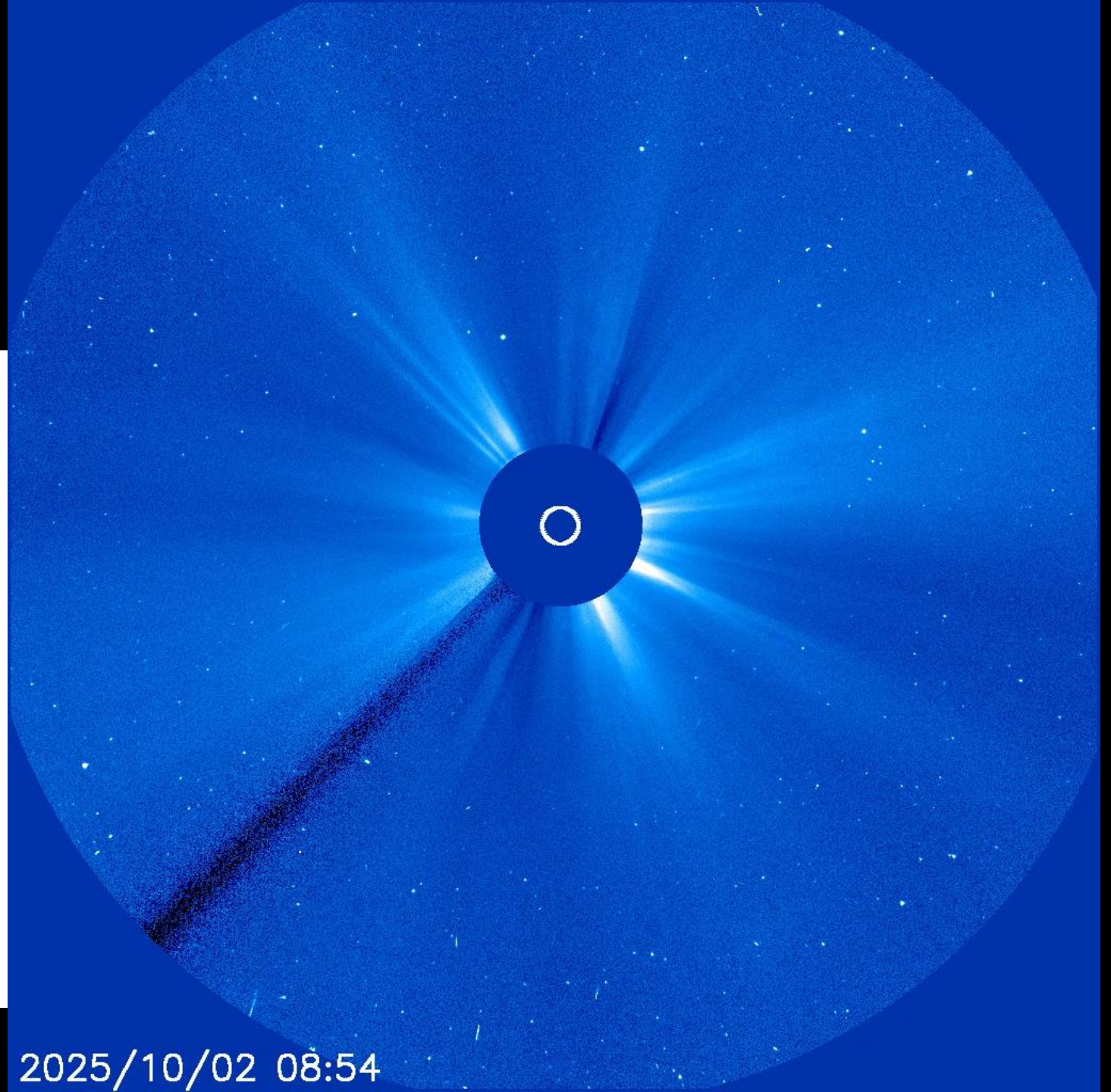


Brighton – April 2025

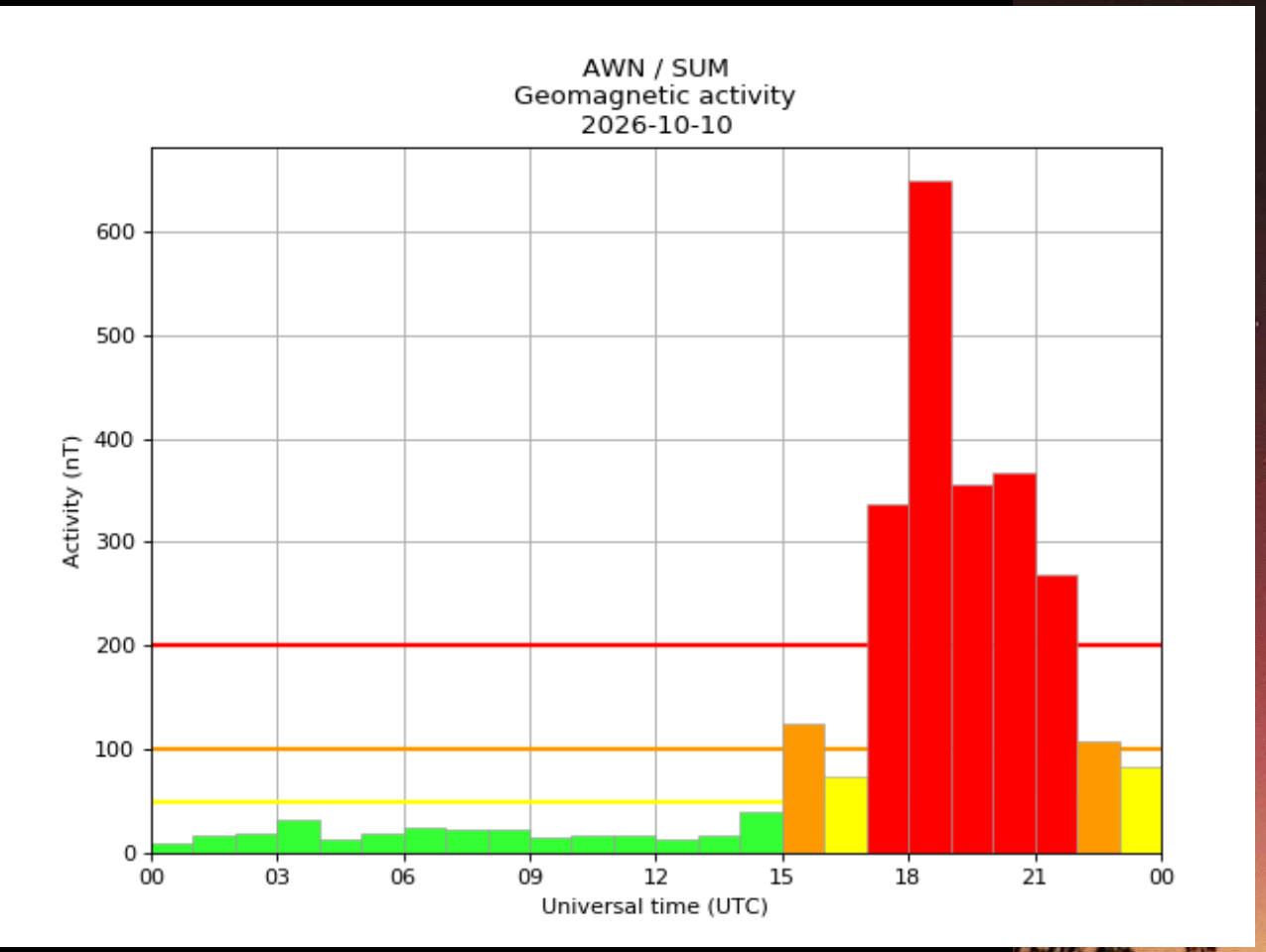


Brighton – October 2025

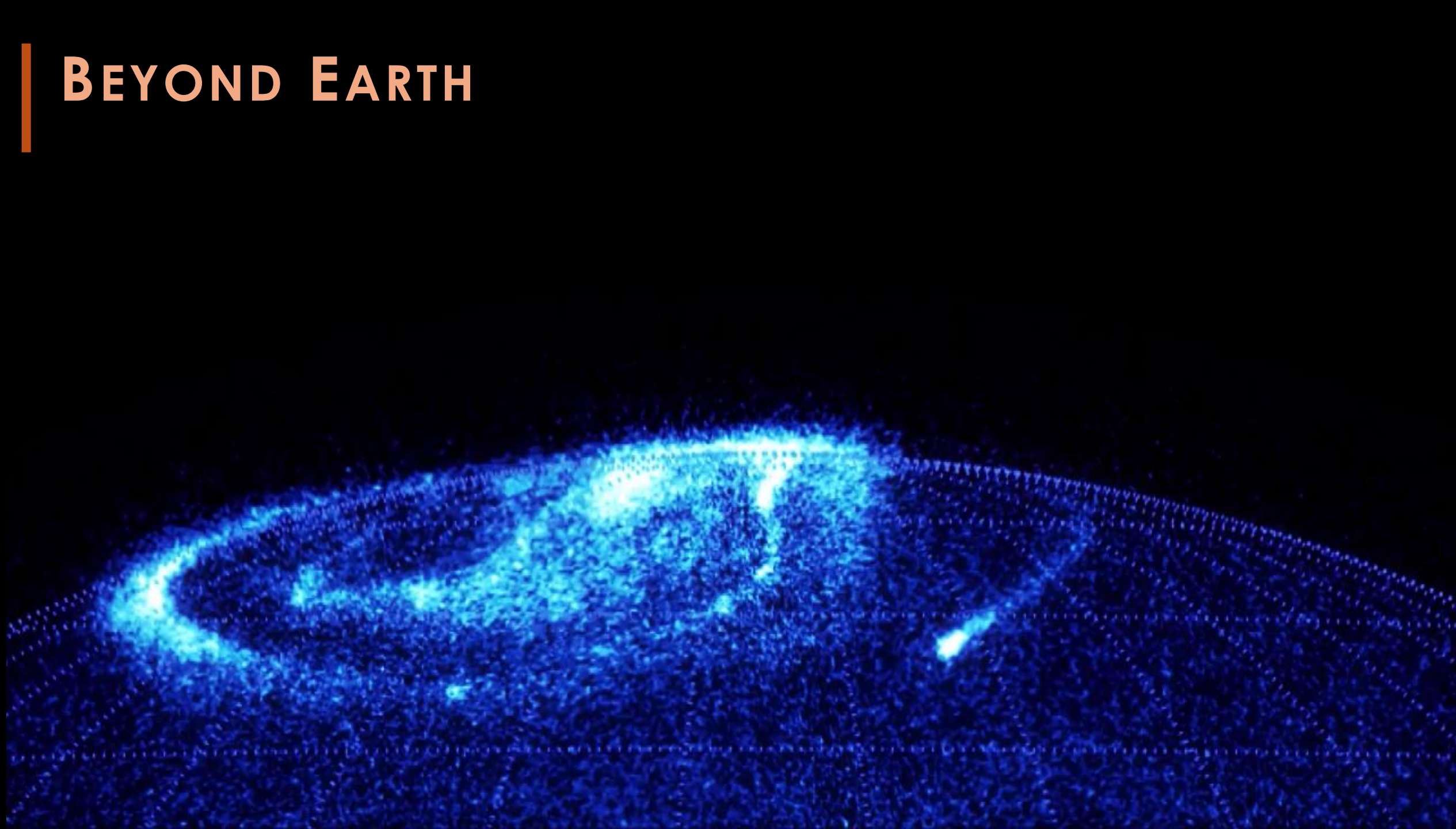




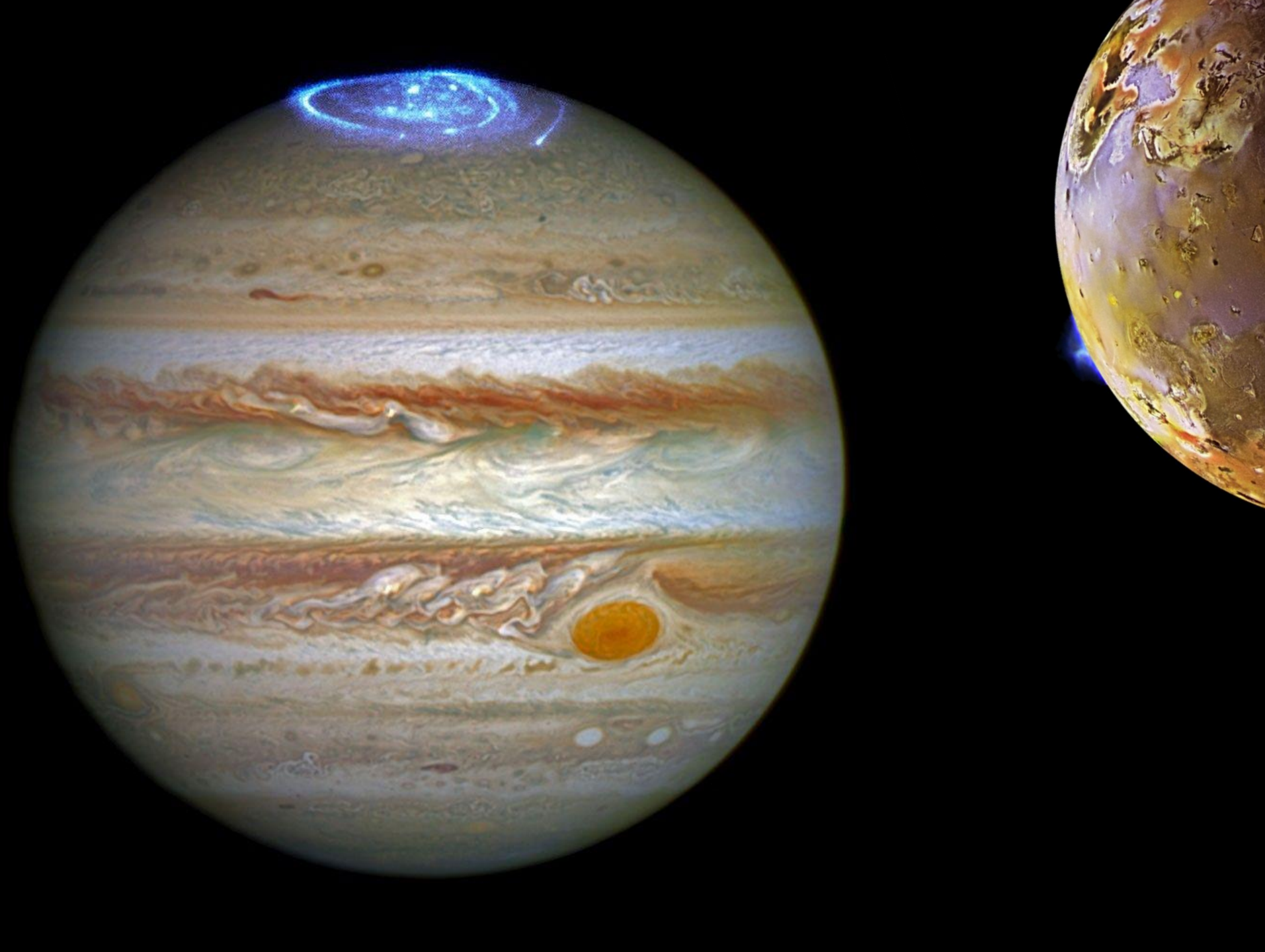
Brighton – October 2026



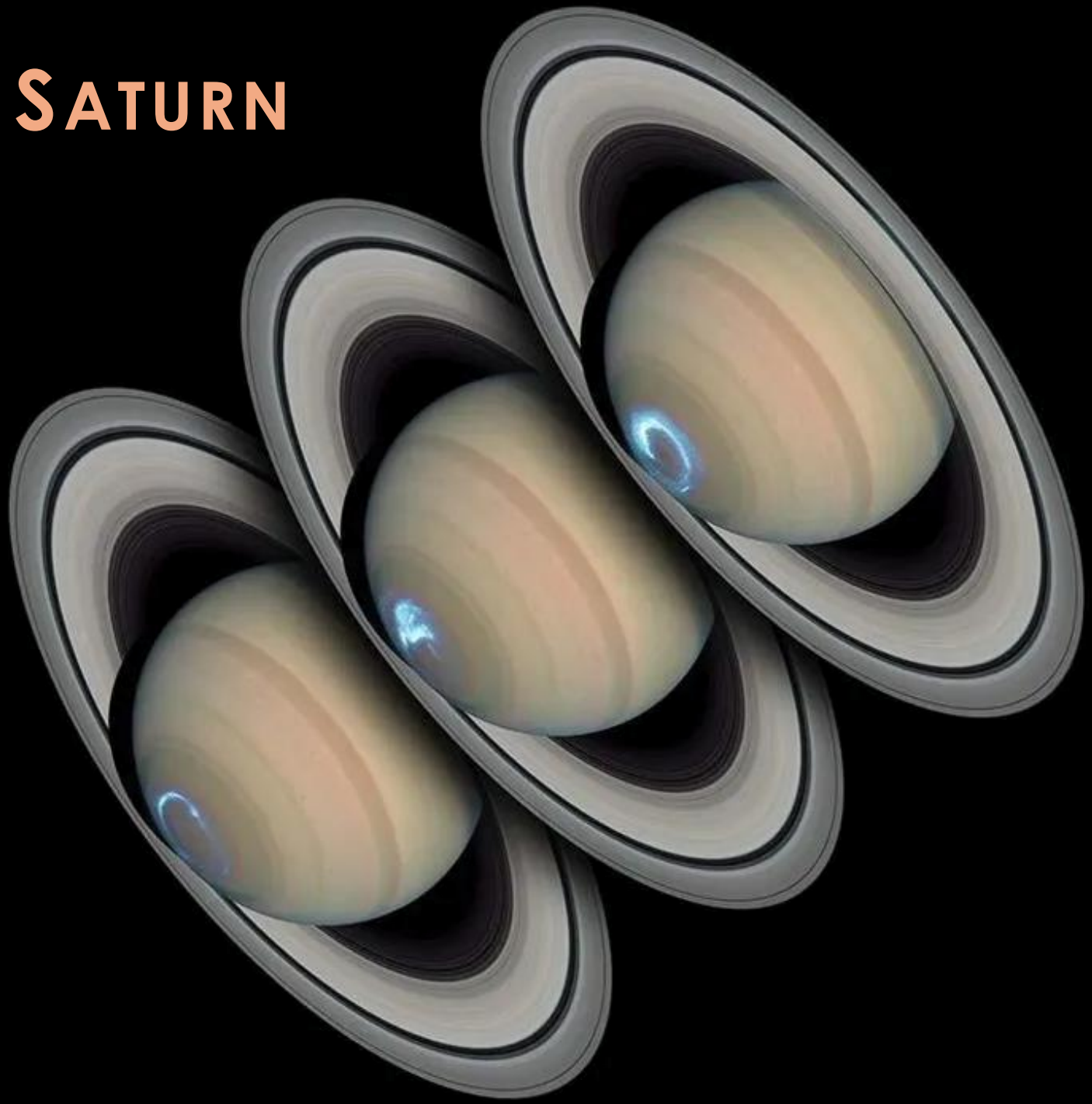
Beyond Earth



Jupiter



SATURN



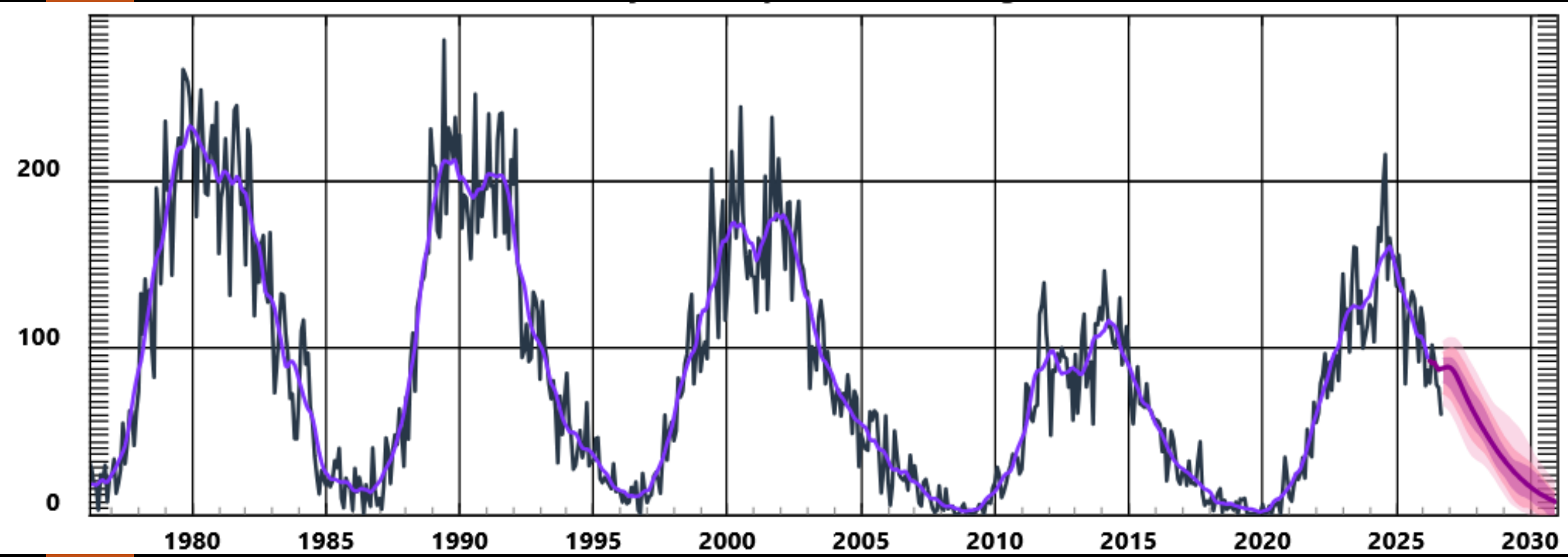
Exoplanets *have even been discovered through the* low-frequency radio-waves ($\lesssim$ 150 MHz) emitted by their aurora.

YZ Ceti
<12ly, 0.7 $M_\oplus$, 0.9$R_\oplus$, 0.02AU [2017]

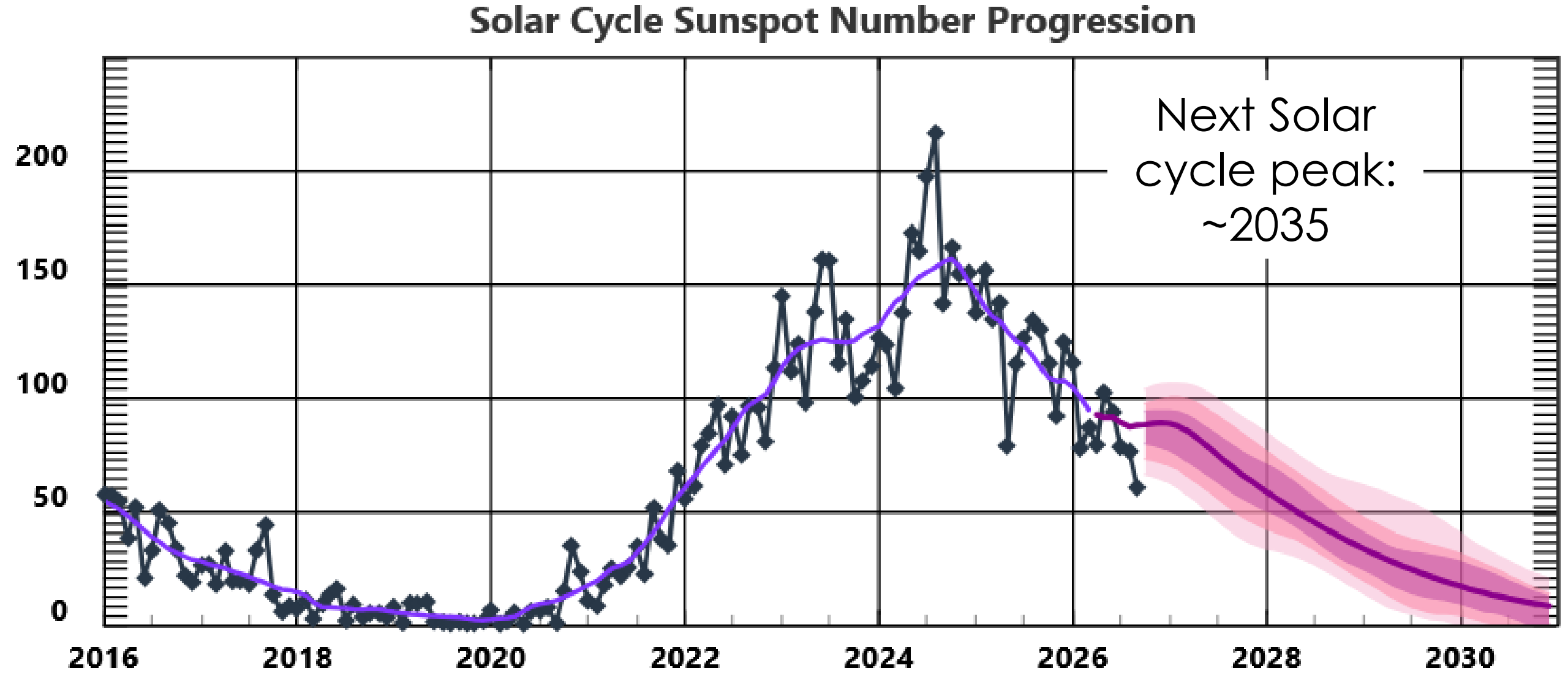
SIMP J0136+0933
20ly, 13 $M_\oplus$, 14 $R_\oplus$ [2003]

GJ 1151
26ly, $\leq 5M_\oplus$, 0.91$R_\oplus$ [2023]

Looking ahead to 2026/7



Looking ahead to 2026/7



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